Supplementary Material S2
MMAT Appraisal by Study and Domain

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder: A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

This document complements the figure in the manuscript presenting the aggregate distribution of MMAT judgements by domain (Section 2.6). Here the appraisal is reported study by study. In accordance with the manual of the Mixed Methods Appraisal Tool (MMAT) version 2018, which deliberately eliminated the numerical summative scoring present in earlier versions, no aggregate score per study is computed or reported. Each study was assigned to the corresponding design category and appraised with the five criteria proper to that category: criteria 2.1–2.5 for randomised controlled trials (Cat. 2), criteria 3.1–3.5 for quantitative nonrandomised studies (Cat. 3), criteria 4.1–4.5 for quantitative descriptive studies (Cat. 4), and criteria 5.1–5.5 for mixed-methods studies (Cat. 5).

The five judgement columns correspond to the five aggregated domains described in Section 2.6 of the manuscript: **Rep.** = Representativeness of the sample (3.1/4.2; 2.2 baseline comparability for Cat. 2); **Meas.** = Adequacy of measurements (3.2/4.3; 2.4 blinded outcome assessment for Cat. 2); **Int.** = Intervention as intended and sampling strategy (3.5/4.1; 2.5 adherence for Cat. 2); **Conf.** = Control of confounding factors and analysis (3.4/4.5; 2.1 randomisation for Cat. 2); **Out.** = Completeness of outcome data (3.3/4.4; 2.3 for Cat. 2). Judgements are coded as **Y** = Yes, **N** = No, **ND** = Not Determinable. The judgements are also recorded in `data/extraction_master.csv`, from which the aggregate figure is computed.

Section A. Eligible corpus (31 studies)

All 31 eligible studies are assignable to an MMAT design category (1 randomised controlled trial, 27 quantitative nonrandomised, 2 quantitative descriptive, 1 mixed methods). Seventeen studies were added in V3.6 after the reconstruction of the screening (identifiers 29 to 51; Supplementary Material S5); their appraisal is first-reviewer coding pending confirmation by a second reviewer. Reed et al. (2021) and Almeida et al. (2019), appraised in the previous version, no longer belong to the eligible corpus because they do not use or evaluate an automated analysis of the face or gaze (criterion CI(1) made explicit in V3.6). Alsubai (2025) and Akın-Bülbül et al. (2024) are documented as contextual evidence in Section B, together with the five studies moved by the audit of criterion CI(3) in V3.6: Rudovic et al. (2018), Dapogny et al. (2019), Rudovic et al. (2019), Sholikah et al. (2024) and Sariyanidi et al. (2025), whose venues had no SJR quartile in the year of publication (`data/sjr_verification.csv`).

#	Reference	Cat.	Rep.	Meas.	Int.	Conf.	Out.
Category 2: Randomised controlled trial (1 study)							
39	Voss et al. (2019)	2	Y	Y	N	Y	Y
Category 3: Quantitative nonrandomised (27 studies)							
1	Johnson et al. (2025)	3	Y	Y	Y	ND	Y
5	Shaik et al. (2025)	3	ND	Y	ND	N	Y
8	Radočaj and Martinović (2025)	3	Y	Y	Y	Y	Y
9	Washington et al. (2022)	3	Y	Y	Y	Y	Y

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#	Reference	Cat.	Rep.	Meas.	Int.	Conf.	Out.
13	Hassan et al. (2025)	3	Y	Y	Y	Y	Y
16	Cicceri et al. (2026)	3	Y	Y	Y	ND	Y
19	Alam et al. (2025)	3	ND	Y	ND	N	Y
20	Hwang et al. (2024)	3	Y	Y	Y	Y	Y
21	Aljbori et al. (2025)	3	N	Y	N	N	Y
23	Vedantham et al. (2025)	3	N	Y	ND	N	Y
24	Hosney et al. (2024)	3	N	Y	N	N	Y
28	Leo et al. (2018)	3	Y	Y	Y	Y	Y
29	Kiejdo et al. (2025)	3	ND	Y	Y	Y	Y
30	Wang et al. (2025)	3	Y	Y	Y	Y	Y
31	Kurian and Tripathi (2024)	3	N	ND	ND	ND	ND
32	Manelis-Baram et al. (2025)	3	ND	Y	Y	Y	ND
33	Chen et al. (2024)	3	Y	Y	Y	Y	ND
34	Shi et al. (2022)	3	N	Y	Y	Y	Y
35	Silva et al. (2021)	3	N	Y	Y	ND	ND
36	Alvari et al. (2021)	3	ND	Y	Y	ND	Y
37	Carpenter et al. (2020)	3	ND	Y	Y	ND	ND
38	Pliska et al. (2025)	3	N	Y	Y	Y	ND
40	Fan et al. (2025)	3	N	ND	Y	N	ND
41	Wall et al. (2025)	3	ND	Y	Y	N	N
48	Kalantarian et al. (2019)	3	N	ND	Y	ND	ND
49	Han et al. (2022)	3	ND	Y	Y	ND	ND
51	Bochet et al. (2021)	3	Y	Y	Y	ND	Y
Category 4: Quantitative descriptive (2 studies)							
2	He et al. (2019)	4	Y	Y	Y	Y	Y
45	Carter Leno et al. (2020)	4	Y	Y	Y	ND	ND
Category 5: Mixed methods (1 study)							
4	Lecciso et al. (2021)	5	Y	Y	Y	ND	Y

Section B. Contextual evidence outside the eligible set (14 studies)

The following 14 studies do not form part of the eligible corpus and do not participate in any denominator of the manuscript. Five infringe CE(3) (exclusively non-facial data); three infringe the age criterion (adult participants); six infringe CI(3) (venue without SJR quartile in the year of publication).

#	Reference	Reason	Cat.	Rep.	Meas.	Int.	Conf.	Out.
Category 3: Quantitative nonrandomised (9 studies)								
C1	El Rhatassi et al. (2024)	Adult participants	3	ND	Y	ND	N	Y
C2	El Rhatassi et al. (2025)	Adult participants	3	ND	Y	ND	N	Y
C4	Nagae & Lee (2022)	EDA only (CE3)	3	Y	Y	ND	ND	Y
C7	O'Sullivan et al. (2023)	Speech only (CE3)	3	Y	Y	Y	ND	Y
C8	Alsubai (2025)	Tabular, text and clinical records only (CE3)	3	N	Y	N	N	Y
C10	Rudovic et al. (2018)	No SJR quartile in 2018 (CI3)	3	Y	Y	Y	Y	Y
C12	Rudovic et al. (2019)	Proceedings without quartile (CI3)	3	Y	Y	Y	Y	Y
C13	Sholikah et al. (2024)	Proceedings without quartile (CI3)	3	ND	Y	ND	N	Y
C14	Sariyanidi et al. (2025)	Proceedings not indexed in SJR (CI3)	3	Y	Y	Y	Y	Y
Category 4: Quantitative descriptive (4 studies)								
C5	Papaioannou et al. (2021)	EEG only (CE3)	4	Y	Y	Y	ND	Y
C6	Xu & Ju (2023)	sMRI only (CE3)	4	Y	Y	Y	Y	Y
C9	Akın-Bülbül et al. (2024)	Journal not indexed in SJR (CI3)	4	Y	Y	Y	ND	Y
C11	Dapogny et al. (2019)	No SJR quartile in 2019 (CI3)	4	Y	Y	Y	Y	Y
Outside MMAT scope (1 study)								
C3	Levy et al. (2024)	Systems design; adult participants.						

Notes.

1. The eight retracted or withdrawn records identified during screening, including Güngör et al. (2025), Mahmood et al. (2025), Kumar et al. (2024) and Ahmad et al. (2024), are excluded from the corpus and do not receive MMAT appraisal; all eight are listed in Supplementary Material S5. The internal # column follows the corpus order shared with Supplementary Material S4 (C prefix for contextual evidence) and does not correspond to bibliographic reference numbers. Gaps in the sequence are not numbering errors: numbers were assigned before the final integrity and eligibility verification and are not reused.
2. Levy et al. (2024), classified as systems design, does not correspond to any MMAT design category and belongs to the contextual evidence.
3. ND in Conf. for Johnson et al. (2025), Lecciso et al. (2021) and Cicceri et al. (2026) reflects partial documentation of confounding control.
4. Negative judgements in Rep. and Conf. for Aljbori et al. (2025), Vedantham et al. (2025) and Hosney et al. (2024) are consistent with use of web repository datasets without clinical validation (Table 5 of the manuscript). Kurian and Tripathi (2024) use web videos (SSBD and YouTube) without verified diagnosis and receive N in Rep.
5. Voss et al. (2019) is a randomised controlled trial; the N in Int. reflects the 51% adherence reported by the authors.
6. The MMAT does not capture risks specific to predictive modelling such as data leakage, image-level versus subject-level partitioning, or model calibration. This limitation is documented in Section 2.9 of the manuscript.